Rotary Wing Aerodynamics (AE 667)



Rotor Hover Performance Prediction Methods

Contextual definition of Performance

- Range of **Thrust** for a rotor (stall limits)

- **Power consumption** at a given thrust

Will define **Figure of Merit (FM)** connecting power with thrust

Prediction Theories

- Momentum Theory (MT)
- Blade Element Theory (BET)
- Blade Element Momentum Theory (BEMT)
- Vortex Theory
  - Prescribed wake method
  - Free wake method

Momentum Theory

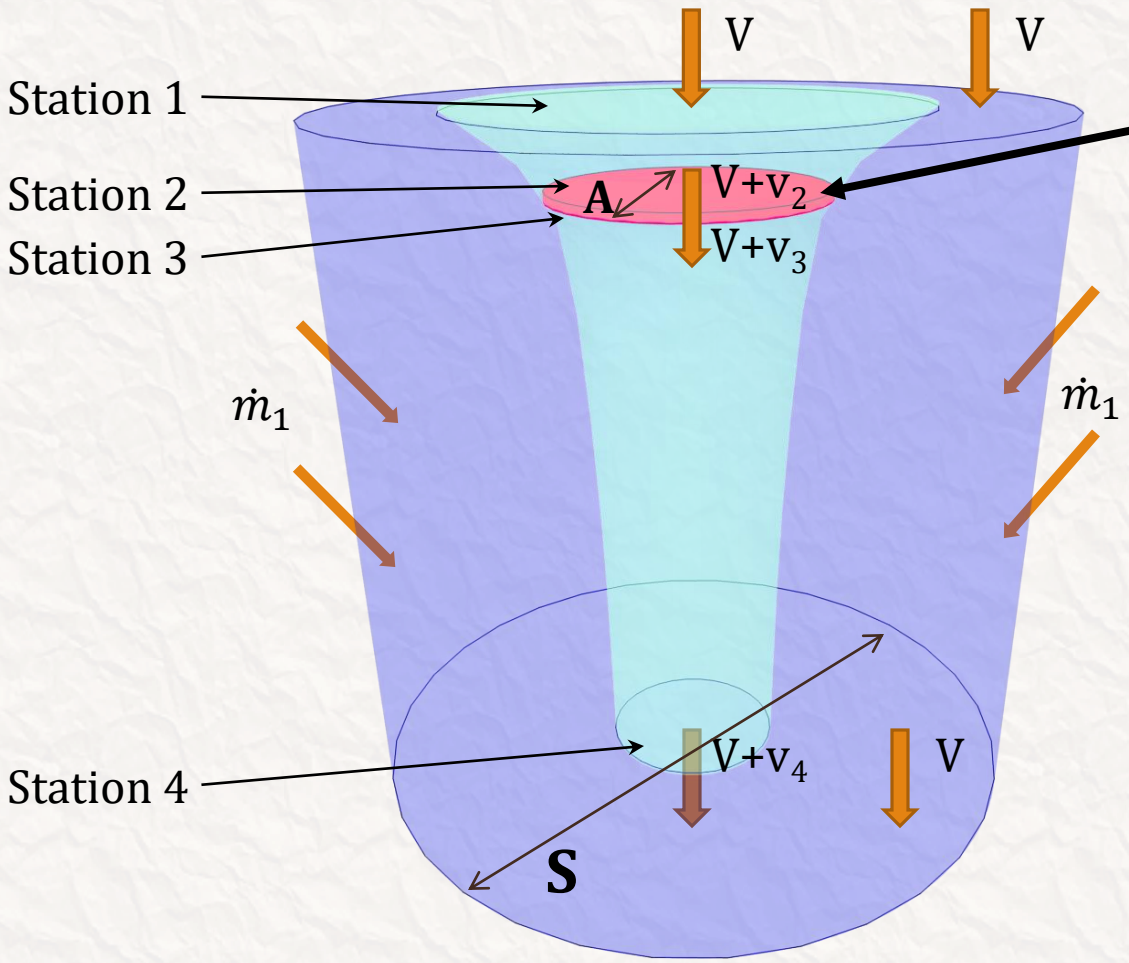
Significance

- Developed in 1865
- Uses simple Mass, Momentum and Energy balance, flow details not covered
- Easy ‘back of an envelop’ type of calculations
- Can predict hover and steady climb (no edgewise component)

Assumptions

- Rotor modeled as an **actuator disk** (no description of blades, number of blades, etc)
- The actuator disk adds momentum to the flow for generating force
- Flow is:
  - **Incompressible, Steady, Inviscid, Irrotational**
- Flow is **1-Dimensional** (totally uniform over the rotor disk)
- **No Swirl**

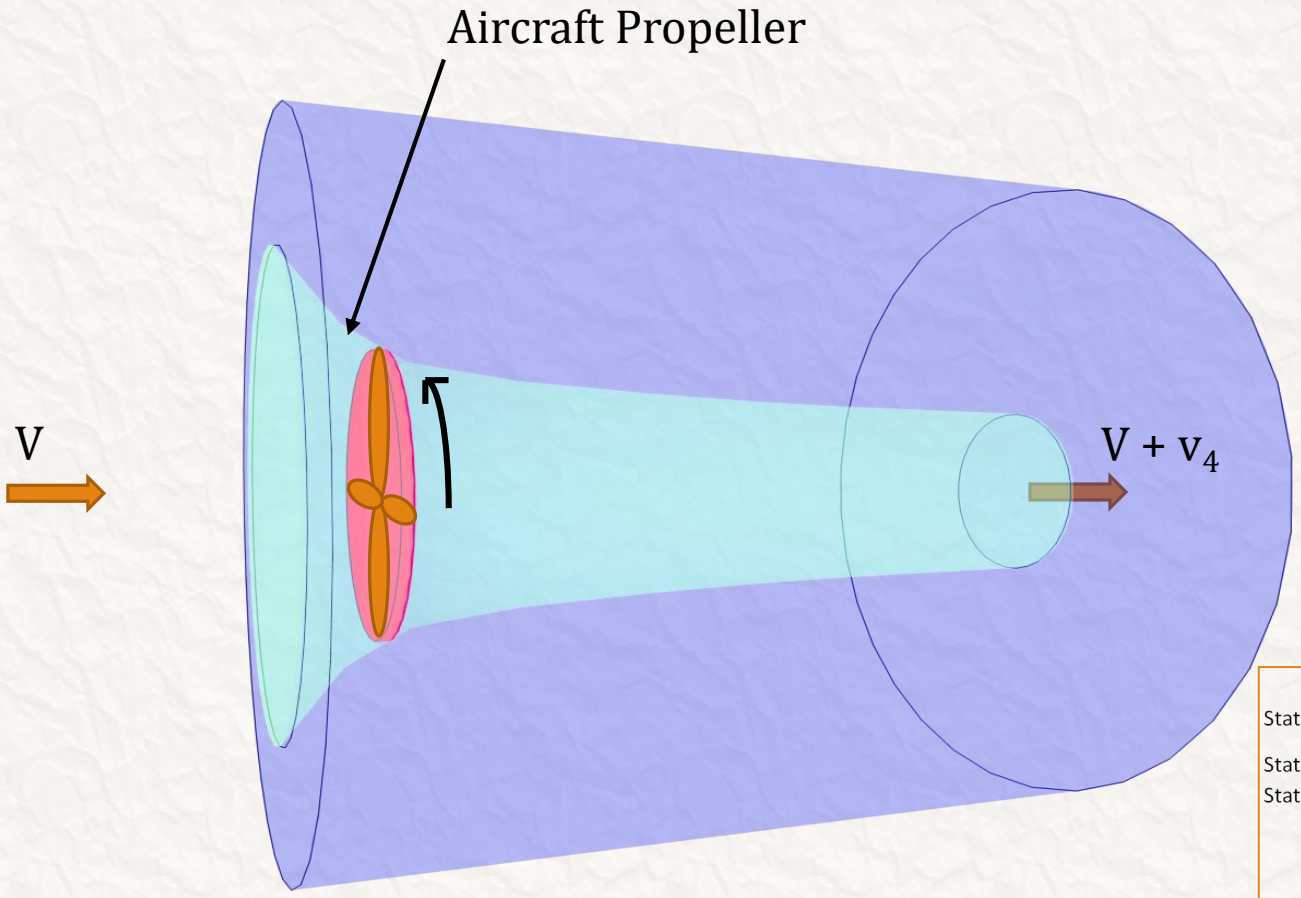
Control Volume

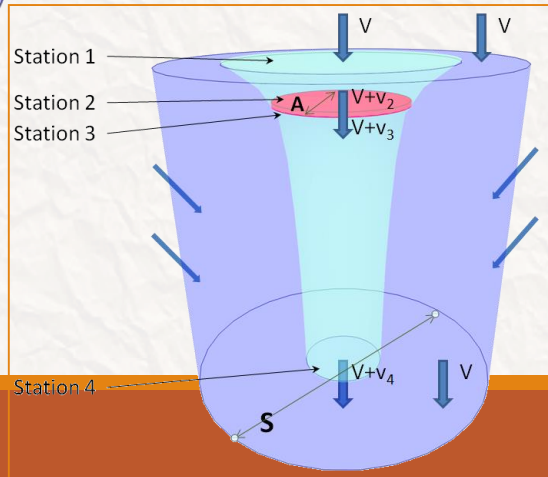


- **V** : Climb velocity
- **v** : Induced velocity
- **A** : Rotor Disk Area
- **S** : Total control volume section area

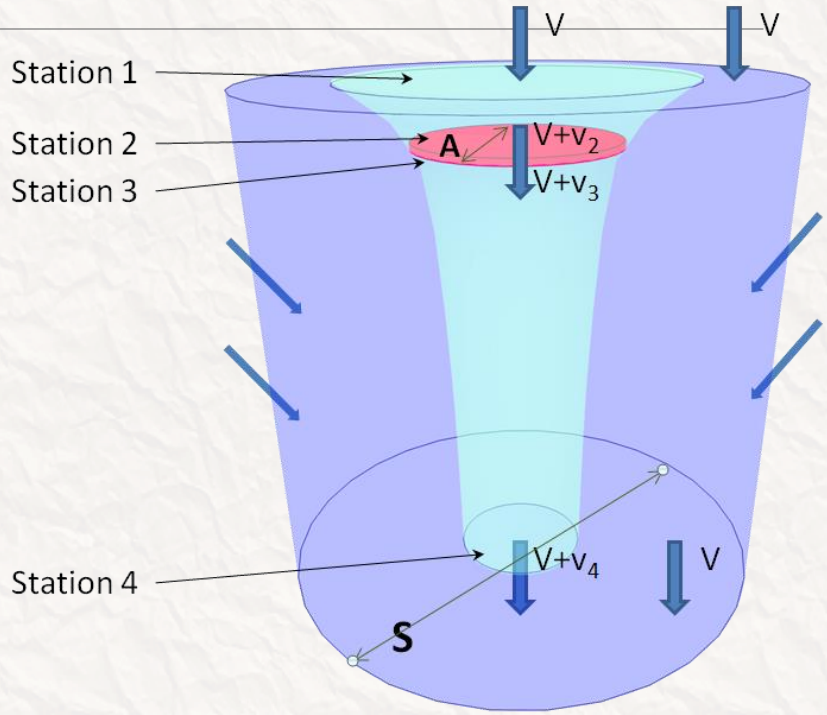
- **Station 1**: Far upstream
- **Station 2**: Just above the rotor
- **Station 3**: Just below the rotor
- **Station 4**: Far downstream

Same thing as a propeller





Mass Conservation



- Mass Inflow
  - through top = ρVS
  - through side = $\dot{m}_1$
- Mass Outflow
  - through bottom

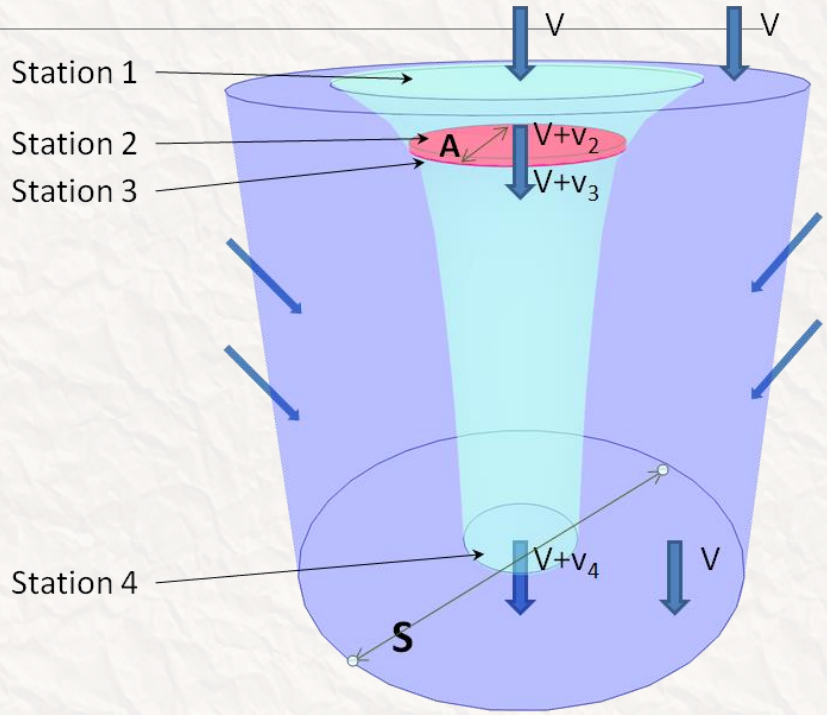
$$\rho V(S - A_4) + \rho(V + v_4)A_4$$

- Inflow = Outflow: $\dot{m}_1 = \rho v_4 A_4$
- Flow through Rotor Disk: $\dot{m} = \rho A(V + v_2) = \rho A(V + v_3)$

$$\dot{m} = \rho A_4(V + v_4)$$

  - Therefore, $v_2 = v_3 = v$

Momentum Conservation (Global)



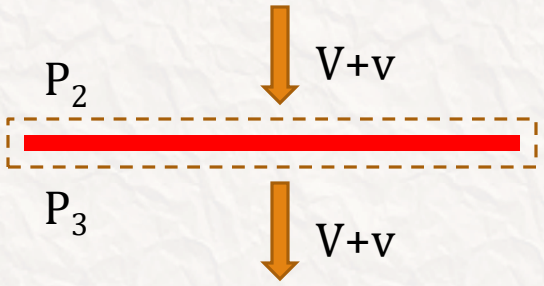
- Momentum Inflow
  - through top = $\rho V^2 S$
  - through side = $\dot{m}_1 V = \rho A_4 v_4 V$
- Momentum Outflow
  - through bottom

$$\rho V^2 (S - A_4) + \rho (V + v_4)^2 A_4$$

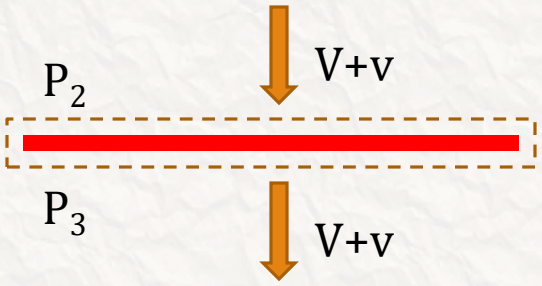
- No pressure difference above and below
- Thrust = Outflow – Inflow: $\boxed{T = \rho A_4 (V + v_4) v_4 = \dot{m} v_4}$

Momentum Conservation (at Disk)

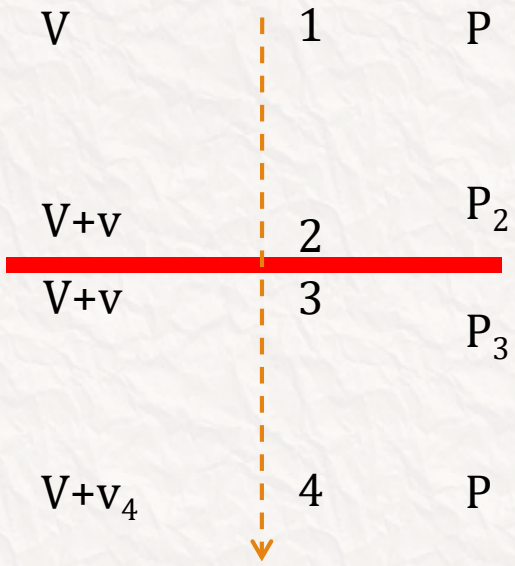
- Momentum Inflow:
  - $\rho(V+v)^2A$
- Momentum Outflow
  - $\rho(V+v)^2A$
- Thrust = $A(P_3 - P_2)$



At Disk



Global



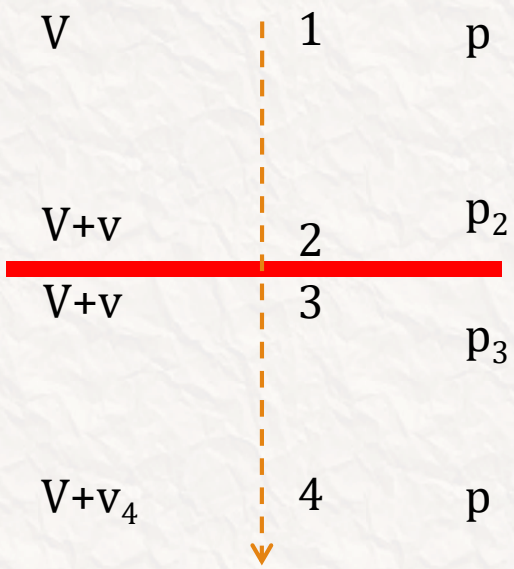
Energy Conservation

- On a particle going through the disk, Bernoulli can be applied between 1-2 & 3-4 (Why?)

$$p_2 + \frac{1}{2}\rho(V+\mathrm{v})^2 = p_\infty + \frac{1}{2}\rho V^2$$

$$p_3 + \frac{1}{2}\rho(V+\mathrm{v})^2 = p_\infty + \frac{1}{2}\rho(V+\mathrm{v}_4)^2$$

$$p_3 - p_2 = \rho\left(V + \frac{\mathrm{v}_4}{2}\right)\mathrm{v}_4$$



Combining

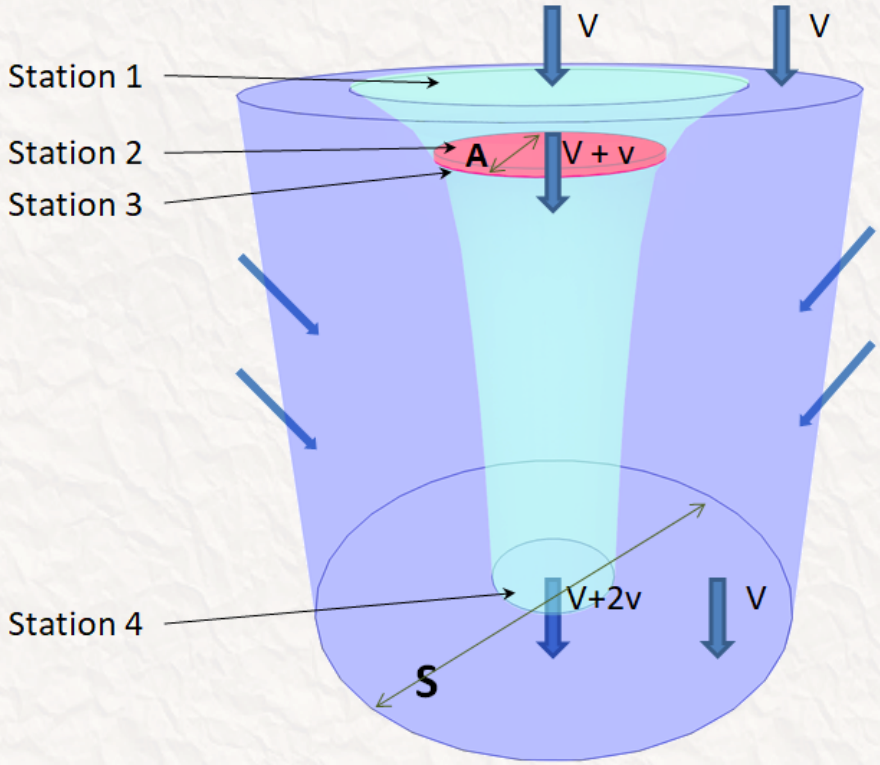
$$p_3 - p_2 = \rho\left(V + \frac{v_4}{2}\right)v_4$$

$$\therefore T = A(p_3 - p_2)$$

$$= \rho A\left(V + \frac{v_4}{2}\right)v_4$$

Also, $T = \rho A(V + v)v_4$

Hence: $v = v_4/2$



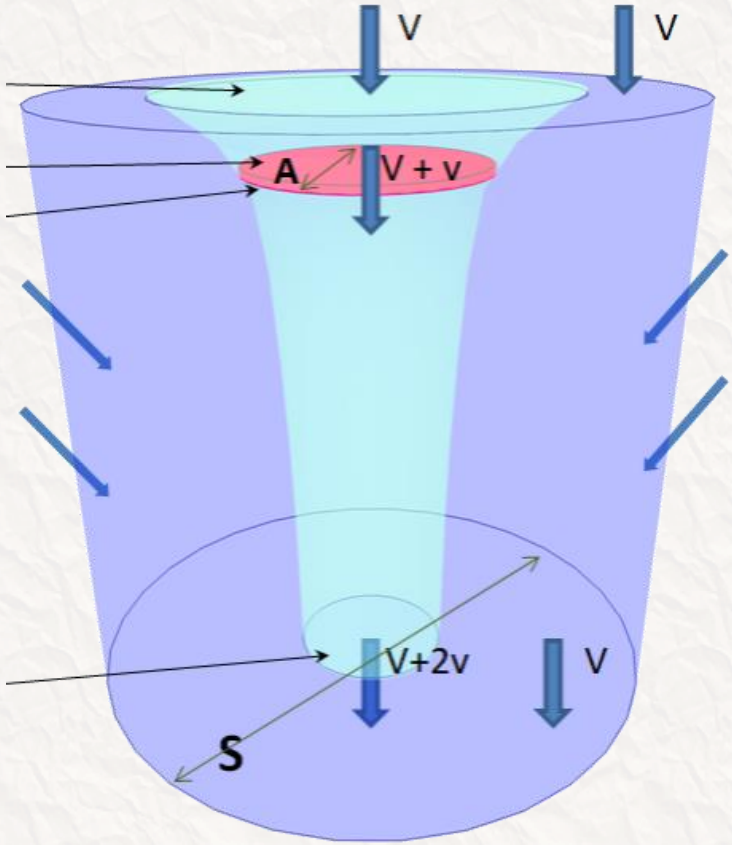
Finding Induced Velocity

$$T = \rho A(V + v)v_4 = 2\rho A(V + v)v$$

$$v = -\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2\rho A}}$$

In Hover, climb velocity $V = 0$

$$v = \sqrt{\frac{T}{2\rho A}}$$



Ideal Power Consumption

$$P = \text{Energy flow out - Energy flow in}$$

$$= \frac{1}{2}\dot{m}\left(\mathrm{V} + 2\mathrm{v}\right)^2 - \frac{1}{2}\dot{m}\mathrm{V}^2$$

$$= 2\dot{m}\mathrm{v}\left(V + \mathrm{v}\right) = T\left(V + \mathrm{v}\right)$$

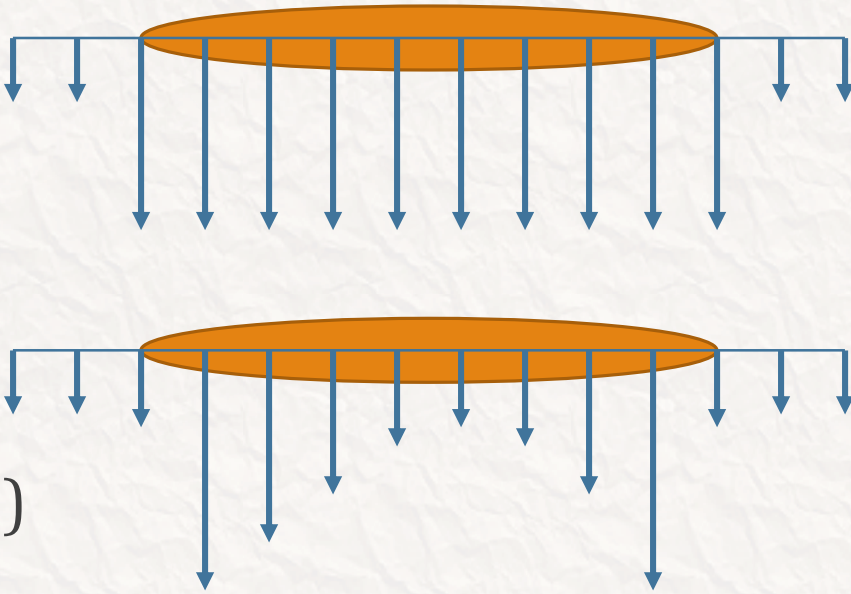
$$= T\left[\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2\rho A}}\right]$$

$$\text{In hover, ideal power} = T\sqrt{\frac{T}{2\rho A}} = Tv$$

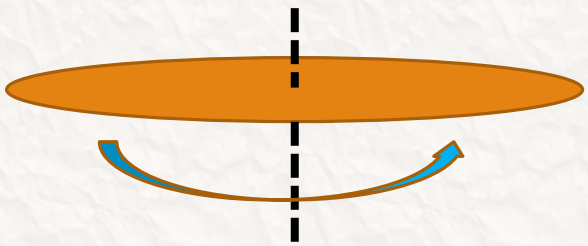
Which losses did we ignore?

Real power requirement to be more because of:

- Non-uniformity of flow
- Tip losses
- Viscous losses
- Swirl losses
- Compressibility effects (shocks)



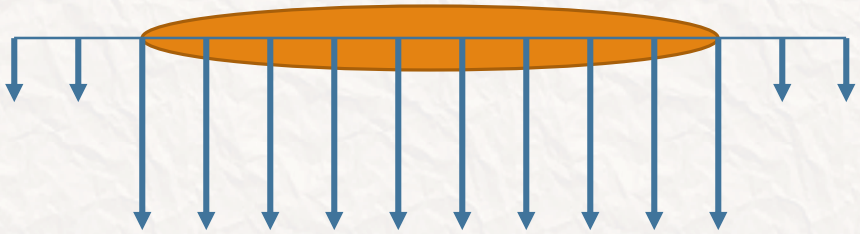




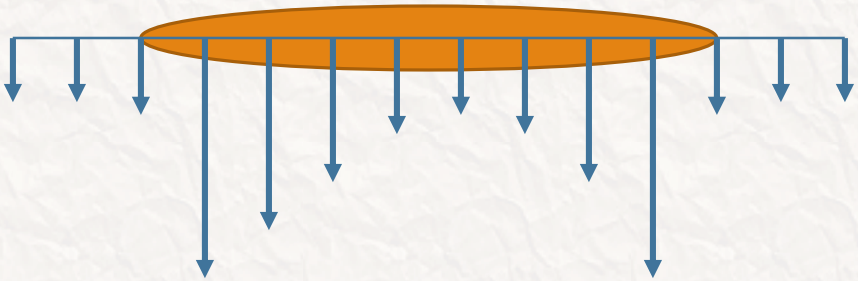
After-Class Activity

Demonstrate by a numerical example that non-uniformity of flow through the rotor leads to more power consumption for a given rotor disk area and thrust demand

Uniform Flow Case
A, T, P_U



Non-Uniform Flow Case
A, T, P_{NU}



Non-Dimensional Forms

- C_T = Thrust Coefficient = $\frac{T}{\rho A(\Omega R)^2}$
- C_P = Power Coefficient = $\frac{P}{\rho A(\Omega R)^3}$
- C_Q = Torque Coefficient = $\frac{Q}{\rho AR(\Omega R)^2}$

In hover, Power = Angular velocity x Torque

$$P = \Omega Q$$

$$C_P = C_Q$$

Figure of Merit

$$FM = \frac{\text{Ideal Power in Hover}}{\text{Actual Power in Hover}}$$

$$= \frac{T\text{v}}{\text{P}} = \frac{C_T \sqrt{\frac{C_T}{2}}}{C_P}$$

Defined for **Hover** case only
(Hence a measure of hover performance only)

In the range of 0.65 – 0.8 for Helicopter rotors

Non-Dimensional Forms

$$\lambda_i = \text{Induced inflow} = \frac{v}{\Omega R} = \frac{1}{\Omega R}\sqrt{\frac{T}{2\rho A}} = \sqrt{\frac{C_T}{2}}$$

$$\text{FM} = \text{Figure of Merit} = \frac{Tv}{P} = \frac{C_T\sqrt{\frac{C_T}{2}}}{C_P}$$

Other terms

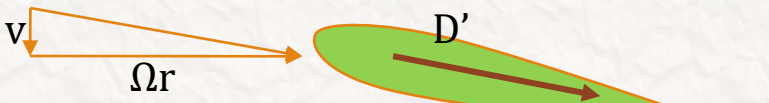
- Power Loading = Thrust / Power
  - Pure helicopters: 3.7 – 6.1 kgf/kW
  - Tilt rotors: 1.2 – 3.7 kgf/kW
  - VTOL fighter jets: < 1.2 kgf/kW

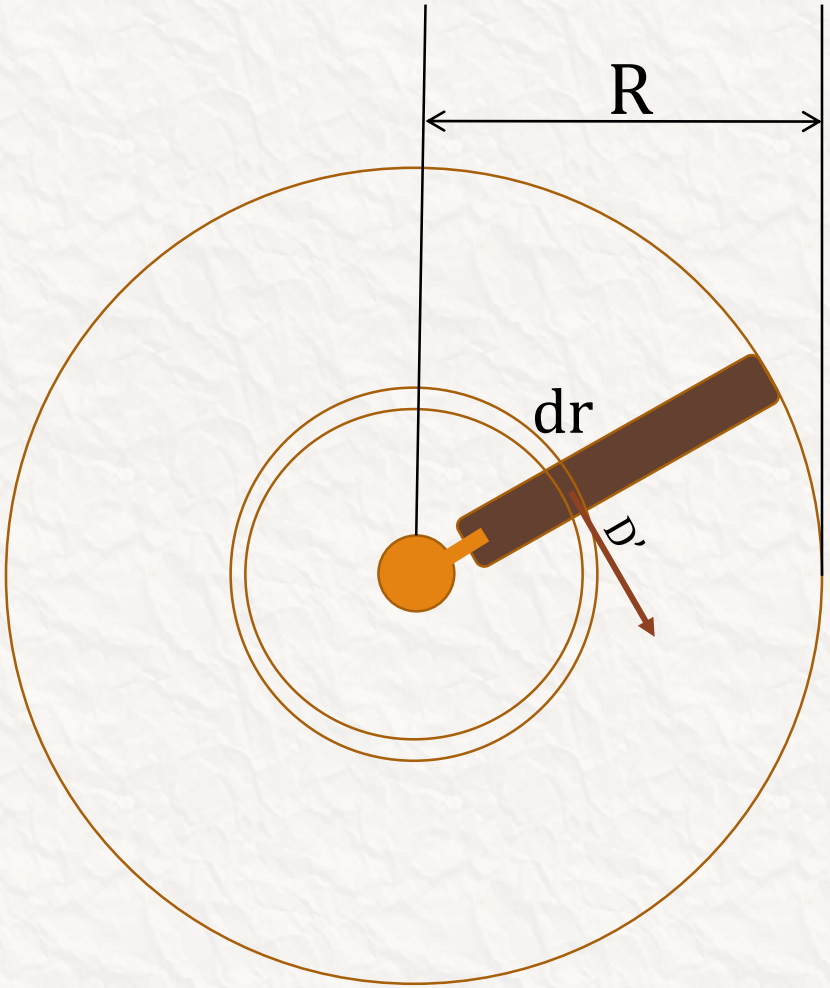


- Disk Loading = Thrust / Disk Area
  - Pure helicopters: 4.8 – 9.6 kgf/m^2
  - Tilt rotors: 19.2 – 38.4 kgf/m^2
  - VTOL fighter jets: in the order of 2500 kgf/m^2



Accounting for viscous losses





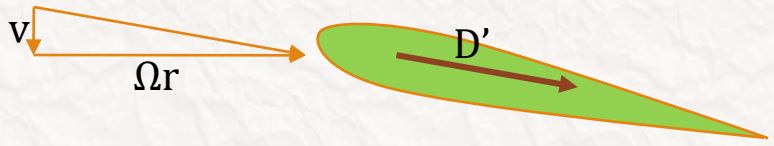
Accounting for viscous losses

- D’ = Drag force per unit blade span (N/m)

  = $\frac{1}{2} C_d \rho \left((\Omega r)^2 + v^2 \right) c \approx \frac{1}{2} C_d \rho (\Omega r)^2 c$

- Power consumed by drag force at a **unit blade section** at radius r = D’(Ωr)

  [neglecting v over Ωr]



Accounting for viscous losses

- Overall viscous drag power loss

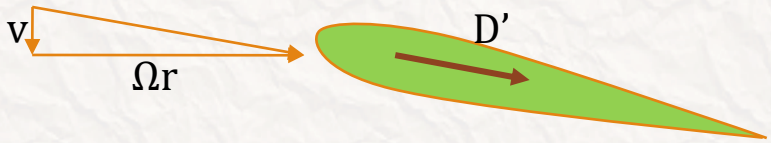
$$P_{Profile} = b \int_{root}^{tip} D'(\Omega r)dr \quad \text{(here } b = \text{ number of blades)}$$

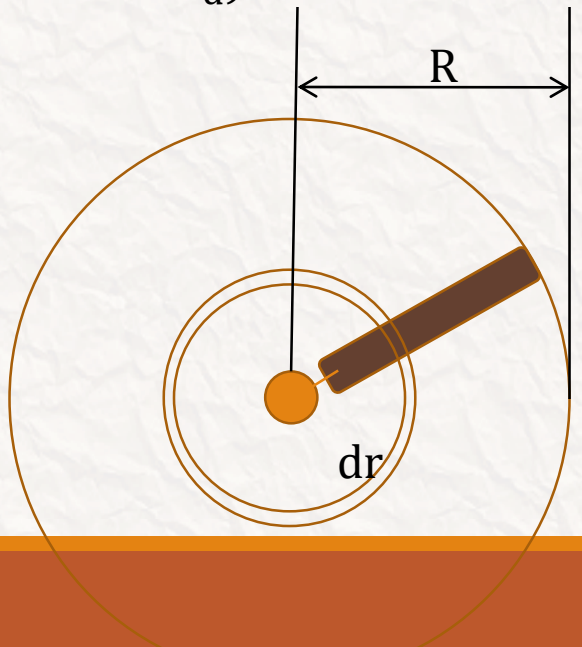
$$= b \int_{0}^{R} \frac{1}{2}\rho c C_d (\Omega r)^2 \Omega r dr$$

$$\cong \frac{bc}{8}\rho C_d (\Omega R)^3 R \quad \text{(for rectangular blade, constant value of } C_d)$$

$$C_{P,\mathrm{Pr}\,ofile} = C_{P,0} = \frac{P_{\mathrm{Pr}\,ofile}}{\rho A(\Omega R)^3} = \frac{bcR}{\pi R^2}\frac{C_d}{8} = \sigma\frac{C_d}{8}$$

where σ = Solidity = Blade area / Disk



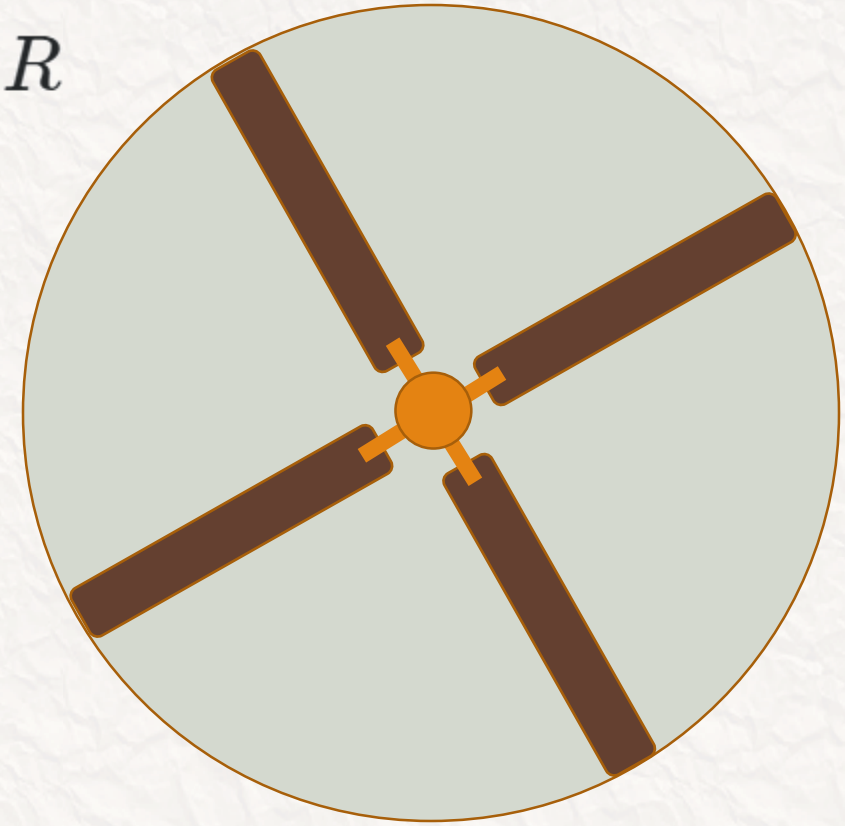


Solidity (σ)

Area of rotor blades = $b \cdot c \cdot R$

Area of rotor disk = $\pi \cdot R^2$

$$Rotor\ solidity = \frac{b \cdot c}{\pi \cdot R}$$



Total Power

Total power = Induced Power + Profile Power

$$C_{P,induced} = C_T\sqrt{\frac{C_T}{2}}; \quad C_{P,\mathrm{Pr}\,ofile} = \sigma\frac{C_d}{8}$$

$$C_P = C_T\sqrt{\frac{C_T}{2}} + \sigma\frac{C_d}{8}$$

Accounting for other sources of power losses through factor k

$$C_P = \kappa C_T\sqrt{\frac{C_T}{2}} + \sigma\frac{C_d}{8} \text{ where } k \approx 1.15$$

Limitations of Momentum Theory

- Not sufficient to design rotors

- Does not account for:
  - Number of blades
  - Airfoil characteristics
  - Blade shape (taper, twist, sweep)
  - Compressibility effects